IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING

EXAMINATIONS 2026

WAVELETS AND APPLICATIONS SOLUTIONS

SOLUTIONS

1. (a) For the resulting filter to have four taps and two zeros at $\omega = \pi$, the filter $G_0(z)$ must satisfy:

$$G_0(z) = \frac{1}{\sqrt{2}}(1 + z^{-1})^2 Q(z)$$

where $Q(z)$ is a first-order polynomial, i.e., $Q(z) = a + bz^{-1}$. [3/10]

To find a and b, we can write the orthogonality condition in z-domain:

$$G_0(z)G_0(z^{-1}) + G_0(-z)G_0(-z^{-1}) = 2$$

and also we can evaluate the orthonogonality condition $\sum_n g_n g_{n-2k} = \delta_k$ for $k = 0$ and $k = 1$. Specifically,

$$\sum_n g_n g_n = 1$$

and

$$\sum_n g_n g_{n-2} = 0.$$

[6/10]

So

$$G_0(z) = \frac{1}{\sqrt{2}}(1+2z^{-1}+z^{-2})(a+bz^{-1}) = \frac{1}{\sqrt{2}}(a+(2a+b)z^{-1}+(2b+a)z^{-2}+bz^{-3}).$$

Consequently, the condition $\sum_n g_n g_n = 1$ yields

$$a^2 + (2a + b)^2 + (2b + a)^2 + b^2 = 2$$

and the condition $\sum_n g_n g_{n-2} = 0$ yields

$$a(2b + a) + (b(2a + b) = 0$$

and this gives the following two conditions $3a^2+3b^2+4ab = 1$ and $a^2+4ab+b^2 = 0$. [8/10]

We also note that the orthogonality condition for $z = 1$ implies that $G_0(1) = \sqrt{2}$ or $a + (2a + b) + (2b + a) + b = 2$ These constraints lead to

$$a^2 + b^2 = \frac{1}{2}$$

and

$$4a + 4b = 2$$

which implies $a = 1/2 - b$. So b is the solution to $(1/2 - b)^2 + b^2 = 1/2$. This is $2b^2 - b - 1/4 = 0$, which yields

$$b = \frac{1 \pm \sqrt{3}}{4}.$$

Both solutions are valid but we pick

$$b = \frac{1-\sqrt{3}}{4}$$

leading to

$$a = \frac{1+\sqrt{3}}{4}$$

and then to the coefficients g_0, g_1, g_2, g_3 as indicated in the exam paper. [10/10]

(b) We have:

$$H_0(z) = G_0(z^{-1}) = \frac{1+\sqrt{3}}{4\sqrt{2}} + \frac{3+\sqrt{3}}{4\sqrt{2}}z + \frac{3-\sqrt{3}}{4\sqrt{2}}z^2 + \frac{1-\sqrt{3}}{4\sqrt{2}}z^3.$$

[2/6]

$G_1(z) = -z^{-1}G_0(-z^{-1})$ ("shift and modulation") so

$$G_1(z) = -z^{-1}G_0(-z^{-1}) = -\frac{1+\sqrt{3}}{4\sqrt{2}}z^{-1} + \frac{3+\sqrt{3}}{4\sqrt{2}} - \frac{3-\sqrt{3}}{4\sqrt{2}}z + \frac{1-\sqrt{3}}{4\sqrt{2}}z^2.$$

[4/6]

Finally,

$$H_1(z) = G_1(z^{-1}) = -\frac{1+\sqrt{3}}{4\sqrt{2}}z + \frac{3+\sqrt{3}}{4\sqrt{2}} - \frac{3-\sqrt{3}}{4\sqrt{2}}z^{-1} + \frac{1-\sqrt{3}}{4\sqrt{2}}z^{-2}.$$

[6/6]

(c) The filter $H_1(z)$ has two zeros at $\omega = 0$ consequently it can annihilate polynomials up to degree one. However, $x[n]$ in this case is of degree two so $y_1[n] \neq 0$. The filter $H_0(z)$ cannot annihilate any polynomial and so also $y_0[n] \neq 0$.

(d) The filter $H_1(z)$ has two zeros at $\omega = 0$ consequently it can annihilate polynomials up to degree one and in this case $x[n]$ is indeed a polynomial of degree one so $y_1[n] = 0$. The filter $H_0(z)$ cannot annihilate any polynomial and so $y_0[n] \neq 0$.

(e) No it is not possible. The minimum length is $L = 6$ because the three zeros at $\omega = \pi$ requirement leads to three constraints and orthogonality adds three more constraints. Consequently we need at least 6 taps to satisfy both conditions and the filter in the question has only 5 taps.

2. (a) The signal is of the form $\sum_n f[n]\varphi(t-n)$ where $\varphi(t)$ is the box function. Therefore:

$$\int_{-\infty}^{\infty} |f(t)|^2 dt = \int_{-\infty}^{\infty} f^2(t) dt = \int_{-\infty}^{\infty} \sum_n \sum_m f[n] f[m] \varphi(t-n)\varphi(t-m) dt \qquad (1)$$

where the first equality is due to the fact that $f(t)$ is a real valued signal. [2/5]

We then have:

$$\int_{-\infty}^{\infty} \sum_n \sum_m f[n] f[m] \varphi(t-n)\varphi(t-m) dt = \sum_n \sum_m f[n] f[m] \int_{-\infty}^{\infty} \varphi(t-n)\varphi(t-m) dt.$$

Since the box function form an orthonormal basis:

$$\int_{-\infty}^{\infty} \varphi(t-n)\varphi(t-m) dt = \delta_{m-n},$$

we have

$$\sum_n \sum_m f[n] f[m] \int_{-\infty}^{\infty} \varphi(t-n)\varphi(t-m) dt = \sum_n f^2[n]. \qquad (2)$$

[4/5]

Combining Eq. (1) with Eq. (2) yields

$$\int_{-\infty}^{\infty} |f(t)|^2 dt = \sum_n f^2[n] = \sum_n |f[n]|^2.$$

Hence, if $f(t) \in L_2$ then $f[n] \in \ell_2$ and viceversa. [5/5]

(b) i. Given the function:

$$f(t) = 1 - \cos(2\pi t), \quad t \in [0, 1]$$

we want to find the orthogonal projection $\hat{f}(t)$ of $f(t)$ onto the subspace V spanned by the orthonormal basis:

$$\{\phi_1, \phi_2, \phi_3, \phi_4\} = \{\varphi(t), \psi(t), \sqrt{2}\psi(2t), \sqrt{2}\psi(2t-1)\}$$

where $\varphi(t)$ is the Haar scaling function and $\psi(t)$ is the Haar wavelet. Since the functions $\varphi(t)$, $\psi(t)$, $\sqrt{2}\psi(2t)$ and $\sqrt{2}\psi(2t-1)$ are orthonormal, we don't need to find the dual basis: $\tilde{\varphi}_i(t) = \varphi_i(t)$.
The coefficients for an orthogonal projection onto an orthonormal basis are given by the inner product $c_i = \langle f, \phi_i \rangle = \int_0^1 f(t)\phi_i(t)\, dt$. [1/5]

Specifically, $\varphi(t) = 1$ for $t \in [0, 1)$ and

$$c_1 = \int_0^1 (1 - \cos(2\pi t)) \cdot 1 \, dt = \left[t - \frac{1}{2\pi} \sin(2\pi t) \right]_0^1 = (1-0) - (0-0) = 1$$

[2/5]

$\psi(t) = 1$ on $[0, 1/2)$ and -1 on $[1/2, 1)$ so

$$c_2 = \int_0^{1/2} (1 - \cos(2\pi t))\, dt - \int_{1/2}^{1} (1 - \cos(2\pi t))\, dt$$

Calculating the first integral: $\left[t - \frac{\sin(2\pi t)}{2\pi}\right]_0^{1/2} = \frac{1}{2}$.
Calculating the second integral: $\left[t - \frac{\sin(2\pi t)}{2\pi}\right]_{1/2}^{1} = 1 - \frac{1}{2} = \frac{1}{2}$.

$$c_2 = \frac{1}{2} - \frac{1}{2} = 0.$$

[3/5]

$\psi_{1,0}(t) = \sqrt{2}$ on $[0, 1/4)$ and $-\sqrt{2}$ on $[1/4, 1/2)$ so

$$c_3 = \sqrt{2}\left(\int_0^{1/4} (1 - \cos(2\pi t))\, dt - \int_{1/4}^{1/2} (1 - \cos(2\pi t))\, dt\right)$$

$$\int_0^{1/4} (1 - \cos(2\pi t))\, dt = \left[t - \frac{\sin(2\pi t)}{2\pi}\right]_0^{1/4} = \frac{1}{4} - \frac{1}{2\pi}$$

$$\int_{1/4}^{1/2} (1 - \cos(2\pi t))\, dt = \left[t - \frac{\sin(2\pi t)}{2\pi}\right]_{1/4}^{1/2} = \frac{1}{2} - \left(\frac{1}{4} - \frac{1}{2\pi}\right) = \frac{1}{4} + \frac{1}{2\pi}.$$

Finally,

$$c_3 = \sqrt{2}\left(\frac{1}{4} - \frac{1}{2\pi} - \left(\frac{1}{4} + \frac{1}{2\pi}\right)\right) = \sqrt{2}\left(-\frac{2}{2\pi}\right) = -\frac{\sqrt{2}}{\pi}.$$

[4/5]

Finally, $\psi_{1,1}(t) = \sqrt{2}$ on $[1/2, 3/4)$ and $-\sqrt{2}$ on $[3/4, 1)$.

$$c_4 = \sqrt{2}\left(\int_{1/2}^{3/4} (1 - \cos(2\pi t))\, dt - \int_{3/4}^{1} (1 - \cos(2\pi t))\, dt\right)$$

Using the symmetry of the cosine function:

$$\int_{1/2}^{3/4} (1 - \cos(2\pi t))\, dt = \left[t - \frac{\sin(2\pi t)}{2\pi}\right]_{1/2}^{3/4} = \left(\frac{3}{4} - \frac{-1}{2\pi}\right) - \frac{1}{2} = \frac{1}{4} + \frac{1}{2\pi}$$

$$\int_{3/4}^{1} (1 - \cos(2\pi t))\, dt = \left[t - \frac{\sin(2\pi t)}{2\pi}\right]_{3/4}^{1} = 1 - \left(\frac{3}{4} + \frac{1}{2\pi}\right) = \frac{1}{4} - \frac{1}{2\pi}$$

$$c_4 = \sqrt{2}\left(\frac{1}{4} + \frac{1}{2\pi} - \left(\frac{1}{4} - \frac{1}{2\pi}\right)\right) = \frac{\sqrt{2}}{\pi}.$$

[5/5]

In conclusion

$$c_1 = 1, \quad c_2 = 0, \quad c_3 = -\frac{\sqrt{2}}{\pi}, \quad c_4 = \frac{\sqrt{2}}{\pi}.$$

ii. The projected function is $\hat{f}(t) = \sum_{i=1}^{4} c_i \phi_i(t)$. Since the space is the Haar space at resolution $j = 2$, $\hat{f}(t)$ is piecewise constant over four equal intervals of length $1/4$.

[1/5]

Calculating the values on each interval $[t_k, t_{k+1}]$:

- **Interval** $[0, 1/4)$**:** $c_1(1) + c_2(1) + c_3(\sqrt{2}) = 1 + 0 - \frac{2}{\pi} = 1 - \frac{2}{\pi} \approx 0.363$
- **Interval** $[1/4, 1/2)$**:** $c_1(1) + c_2(1) + c_3(-\sqrt{2}) = 1 + 0 + \frac{2}{\pi} = 1 + \frac{2}{\pi} \approx 1.637$
- **Interval** $[1/2, 3/4)$**:** $c_1(1) + c_2(-1) + c_4(\sqrt{2}) = 1 + 0 + \frac{2}{\pi} = 1 + \frac{2}{\pi} \approx 1.637$
- **Interval** $[3/4, 1)$**:** $c_1(1) + c_2(-1) + c_4(-\sqrt{2}) = 1 + 0 - \frac{2}{\pi} = 1 - \frac{2}{\pi} \approx 0.363$

The function $\hat{f}(t)$ can be defined as:

$$\hat{f}(t) = \begin{cases} 1 - \frac{2}{\pi} & 0 \le t < 1/4 \\ 1 + \frac{2}{\pi} & 1/4 \le t < 3/4 \\ 1 - \frac{2}{\pi} & 3/4 \le t < 1 \end{cases}$$

iii. We need to verify that $\langle e(t), \phi_i(t) \rangle = 0$ for $i = 1, .., 4$ with

$$e(t) = f(t) - \hat{f}(t) = \begin{cases} \frac{2}{\pi} - \cos(2\pi t) & 0 \le t < 1/4 \\ -\frac{2}{\pi} - \cos(2\pi t) & 1/4 \le t < 3/4 \\ \frac{2}{\pi} - \cos(2\pi t) & 3/4 \le t < 1 \end{cases}$$

[1/5]

So

$$\langle e(t), \varphi(t) \rangle = \int_0^1 e(t)dt = 0,$$

$$\langle e(t), \psi(t) \rangle = \int_0^{1/2} e(t)dt - \int_{1/2}^1 e(t)dt = 0 + 0 = 0,$$

$$\langle e(t), \psi_{1,0}(t) \rangle = \sqrt{2} \left(\int_0^{1/4} e(t)dt - \int_{1/4}^{1/2} e(t)dt \right) = 0 + 0 = 0$$

and similarly

$$\langle e(t), \psi_{1,1}(t) \rangle = \sqrt{2} \left(\int_{1/2}^{3/4} e(t)dt - \int_{3/4}^1 e(t)dt \right) = 0.$$

[5/5]

(c) $\varphi(t)$ is a B-spline of order N. Therefore,

$$\hat{\varphi}(\omega) = e^{-jN\omega/2} \left(\frac{\sin \omega/2}{\omega/2} \right)^N.$$

Since $|e^{-iN\omega/2}| = 1$, the squared magnitude is:

$$|\hat{\varphi}(\omega)|^2 = \left(\frac{\sin(\omega/2)}{\omega/2} \right)^{2N}$$

[1/5]

Therefore, the function $A(\omega)$ is given by:

$$A(\omega) = \sum_{k\in\mathbb{Z}} |\hat{\varphi}(\omega + 2\pi k)|^2 = \sum_{k\in\mathbb{Z}} \left(\frac{\sin(\frac{\omega+2\pi k}{2})}{\frac{\omega+2\pi k}{2}} \right)^{2N}.$$

Note that $\sin(\frac{\omega+2\pi k}{2}) = \sin(\frac{\omega}{2} + \pi k)$. Using the periodicity of the sine function:

$$\sin\left(\frac{\omega}{2} + \pi k\right) = (-1)^k \sin\left(\frac{\omega}{2}\right)$$

because the power $2N$ is even, $((-1)^k)^{2N} = 1$, thus:

$$\sin^{2N}\left(\frac{\omega}{2} + \pi k\right) = \sin^{2N}\left(\frac{\omega}{2}\right)$$

This term is independent of k and can be pulled out of the summation. [3/5]

Substituting the simplified sine term back into the sum:

$$A(\omega) = \sin^{2N}\left(\frac{\omega}{2}\right) \sum_{k\in\mathbb{Z}} \frac{1}{\left(\frac{\omega+2\pi k}{2}\right)^{2N}}.$$

Factor out the 2^{2N} from the denominator of the sum:

$$A(\omega) = \sin^{2N}\left(\frac{\omega}{2}\right) \cdot 2^{2N} \sum_{k\in\mathbb{Z}} \frac{1}{(\omega + 2\pi k)^{2N}}.$$

Combining the constants outside the sum, we obtain:

$$A(\omega) = \left(2\sin\frac{\omega}{2}\right)^{2N} \sum_{k\in\mathbb{Z}} (\omega + 2\pi k)^{-2N}.$$

[5/5]

3. (a)

$$\hat{X}(z) = \frac{G_0(z)}{2}\left[H_0(z)X(z) + H_0(-z)X(-z)\right] + \frac{G_1(z)}{2}\left[H_1(z)X(z) + H_1(-z)X(-z)\right].$$

[3/5]

Therefore PR condition is:

$$G_0(z)H_0(z) + G_1(z)H_1(z) = 2$$

[4/5]

and

$$G_0(z)H_0(-z) + G_1(z)H_1(-z) = 0.$$

[5/5]

(b) i. We can find $G_0(z)$ by performing polynomial division. A faster approach is the following. We note that $P(z)$ is symmetric and has 7 taps, moreover $H_0(z)$ has 4 taps and is also symmetric. Therefore $G_0(z)$ is symmetric and has 4 taps. So we can write:

$$P(z) = H_0(z)(a + bz^{-1} + bz^{-2} + az^{-3}).$$

or

$$\frac{1}{16}\left(-1 + 9z^{-2} + 16z^{-3} + 9z^{-4} - z^{-6}\right) = \left(\frac{1+z^{-1}}{2}\right)^3 (a + bz^{-1} + bz^{-2} + az^{-3}).$$

[4/10]

To find a and b we evaluate the equality for $z = 1$ and $z = \infty$. We have:

$$2 = 2(a + b)$$

and

$$-\frac{1}{16} = \frac{a}{8}.$$

This leads to

$$G_0(z) = \left(-\frac{1}{2} + \frac{3}{2}z^{-1} + \frac{3}{2}z^{-2} - \frac{1}{2}z^{-3}\right).$$

[7/10]

This is a biorthogonal filter bank, consequently

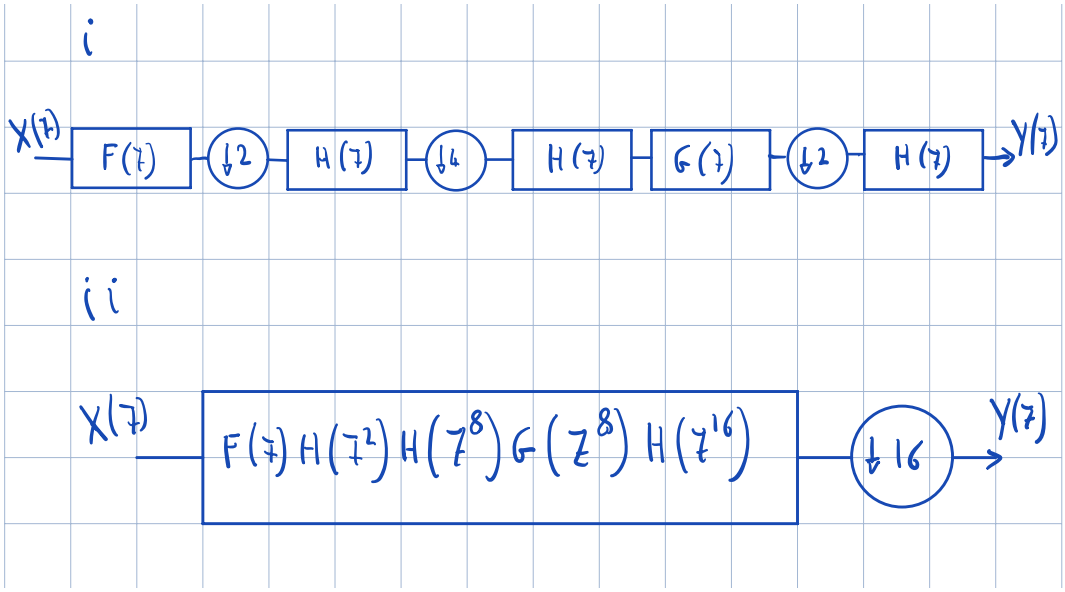
$$H_1(z) = zG_0(-z) = \left(-\frac{1}{2}z - \frac{3}{2} + \frac{3}{2}z^{-1} + \frac{1}{2}z^{-2}\right)$$

and

$$G_1(z) = z^{-1}H_0(-z) = z^{-1}\left(\frac{1-z^{-1}}{2}\right)^3.$$

[10/10]

(c) The equivalent representation is shown in the figure below:



4. (a) Write the polyphase matrix.

$$D(z) = O(z) - P(z)E(z),$$

where

$$P(z) = \beta(1 + z^{-1}).$$

[2/7]

From above equations, we have that:

$$\begin{aligned} S(z) &= E(z) + U(z)(O(z) - P(z)E(z)) \\ &= (1 - U(z)P(z))E(z) + U(z)O(z). \end{aligned}$$

[4/7]

So in matrix form:

$$\begin{bmatrix} S(z) \\ D(z) \end{bmatrix} = \begin{bmatrix} 1 - U(z)P(z) & U(z) \\ -P(z) & 1 \end{bmatrix} \begin{bmatrix} E(z) \\ O(z) \end{bmatrix},$$

which means that the analysis polyphase components are:

$$\begin{aligned} &H_{0e}(z) = 1 - U(z)P(z), H_{0o}(z) = U(z) \\ &H_{1e}(z) = -P(z), H_{1o}(z) = 1. \end{aligned}$$

[7/7]

(b) Write $H_0(z)$ and $H_1(z)$. We have that:

$$H_{1e}(z) = -P(z) = -\frac{1}{2}(1 + z^{-1})$$

and

$$H_{1o}(z) = 1.$$

Therefore,

$$H_1(z) = H_{1e}(z^2) + z^{-1}H_{1o}(z^2) = -\frac{1}{2}(1 + z^{-2}) + z^{-1} = -\frac{1}{2} + z^{-1} - \frac{1}{2}z^{-2}.$$

So the analysis highpass filter is a 3-tap filter $(-\frac{1}{2}, 1, -\frac{1}{2})$.

[3/6]

Then we compute the analysis lowpass filter. First we compute $U(z)P(z)$:

$$U(z)P(z) = \frac{\gamma}{2}(2 + z + z^{-1}),$$

so

$$1 - UP = 1 - \gamma - \frac{\gamma}{2}(z + z^{-1}),$$

which gives

$$H_{0e}(z^2) = 1 - \gamma - \frac{\gamma}{2}(z^2 + z^{-2})$$

hence,

$$H_0(z) = H_{0e}(z^2) + z^{-1}H_{0o}(z^2) = 1 - \gamma - \frac{\gamma}{2}(z^2 + z^{-2}) + \gamma(z + z^{-1}).$$

[5/6]

Since $\gamma = 1$, we have:

$$H_0(z) = H_{0e}(z^2) + z^{-1}H_{0o}(z^2) = -\frac{1}{2}(z^2 + z^{-2}) + (z + z^{-1}).$$

[6/6]

(c) The best is to operate in the time domain. We first note that $P(z) = (1 + z^{-1})/2$. This means that the predictor takes two consecutive elements and compute their average. So, $x_e[n] = 2n$ and the output of the predictor is

$$x_P[n] = (x_e[n] + x_e[n+1])/2 = (2n + 2(n+1))/2 = 2n + 1.$$

[3/6]

So $d[n] = x_o[n] - x_P[n] = 2n + 1 - (2n + 1) = 0$.

[4/6]

Given that $d[n] = 0$ then $s[n] = x_e[n]$. This is not surprising since from the previous answer, we note that the highpass analysis filter $H_1(z) = -\left(1 - 2z^{-1} + z^{-2}\right) = -(1 - z^{-1})^2$. So $H_1(z)$ has double zero at $z = 1$. So the analysis highpass filter has 2 vanishing moments. This means constants and linear polynomials are annihilated.

[6/6]

(d) i. Determine α, β, η such that $d[n] = 0$ for all signals $x[n] = an^2 + b[n] + c$.
In other words, the predictor has to be exact for

$$x[n] = 1, x[n] = n, x[n] = n^2$$

because any quadratic polynomial $an^2 + bn + c$ is a linear combination of these sequences.
We first focus on $x[n] = 1$. Condition $d[n] = 0$ gives: $1 = \alpha + \beta + \eta$.

[2/6]

Then we consider $x[n] = n$. Here $x_e[n] = 2n$ and $x_o[n] = 2n + 1$. Therefore the predictor output is:

$$\alpha x_e[n-1] + \beta x_e[n] + \eta x_e[n+1] = 2n(\alpha + \beta + \eta) + 2(\eta - \alpha).$$

Since $\alpha + \beta + \eta = 1$, the above equation becomes:

$$2n + 2(\eta - \alpha) = 2n + 1 \Rightarrow \eta - \alpha = \frac{1}{2}.$$

[4/6]

Finally we consider $x[n] = n^2$. Here $x_e[n] = 4n^2$ and $x_o[n] = (2n+1)^2$. Therefore, the predictor output becomes:

$$\alpha \cdot 4(n-1)^2 + \beta \cdot 4n^2 + \eta \cdot 4(n+1)^2 = 4n^2(\alpha + \beta + \eta) + 8n(\eta - \alpha) + 4(\alpha + \eta)$$

Using the earlier constraints $\alpha + \beta + \gamma = 1$ and $\eta - \alpha = \frac{1}{2}$, this becomes:

$$4n^2 + 8n \cdot \frac{1}{2} + 4(\alpha + \eta) = 4n^2 + 4n + 4(\alpha + \eta).$$

Exactness requires:

$$4n^2 + 4n + 1 = 4n^2 + 4n + 4(\alpha + \eta) \Rightarrow \alpha + \eta = \frac{1}{4}.$$

Now we solve the two equations:

$$\begin{cases} \eta - \alpha = \frac{1}{2} \\ \alpha + \eta = \frac{1}{4} \end{cases} \Rightarrow \begin{cases} \alpha = -\frac{1}{8} \\ \beta = \frac{3}{4} \end{cases}$$

Then $\beta = 1 - \alpha - \eta = \frac{3}{4}$. So finally, $\alpha = -\frac{1}{8}, \beta = \frac{3}{4}, \eta = \frac{3}{8}$. [6/6]